SPADE · ENGINEERING WORKBENCH

Installation Guide

github.com/lucianoradrigan/SPADE
A complete, step-by-step guide for Windows, Linux and macOS. Every prerequisite is explained (what it is, why it is needed, how to check and install it) and every step lists what you should see and what to do if it fails.

Windows 10 / 11 | Ubuntu 22.04 · 24.04 · 26.04 | macOS Apple Silicon & Intel | No GPU needed

Two ways to install SPADE

Route A · Windows installer (.exe)

For people who just want to **use** SPADE on Windows.

Download one file, double-click, follow the wizard. No Python, no Git, no administrator rights. About 2 GB.

Route B · From the repository (source code)

For Linux and macOS, and for anyone who wants to **modify** SPADE or always run the latest version.

Install the prerequisites, download the code and create a dedicated environment. About 3 GB.

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SECTION 1

Before you start

Read this once. It prevents most installation problems.

1.1 Which route is right for me?

Are you on Windows and only want to use SPADE (not program or change the code)?

Yes → Route A (section 2) | No → keep reading

Which operating system do you have?

Windows → section 3 | Linux (Ubuntu) → section 4 | Mac → section 5

	Route A · .exe installer	Route B · Repository
Systems	Windows 10 / 11, 64-bit only	Windows, Linux and macOS
What to install first	Nothing	Python 3.11 or 3.12 (64-bit), Git and a few system tools
Administrator rights	No	Windows: once (long paths). Linux/Mac: to install system tools.
Difficulty	Very low	Medium: you type commands in a terminal
SPADE version	The one in the latest published installer	The latest code; updated with two commands
Can modify the code	No (not intended for that)	Yes

1.2 Check your computer

Do these checks before installing anything. Each row says how to check and what to do if your computer does not meet the requirement.

Requirement	How to check	If it is not met
64-bit operating system	**Windows:** Start → Settings → System → *About* → "System type" says *64-bit operating system*. **Ubuntu:** `uname -m` prints `x86_64`. **Mac:** every supported Mac is 64-bit.	SPADE cannot run on a 32-bit system (TensorFlow only exists for 64-bit). Use another computer.
System version	**Windows:** press `Win` + `R`, type `winver`, Enter: must be Windows 10 version 21H2 or later, or Windows 11. **Ubuntu:** `lsb_release -a` → 22.04, 24.04 or 26.04. **Mac:** Apple menu → *About This Mac*.	Update Windows (Settings → Windows Update). Other Linux distributions may work but are not covered by this guide.

Free disk space	≈3 GB for Route B, ≈2 GB for Route A. **Windows:** File Explorer → *This PC*. **Ubuntu/Mac:** `df -h ~` (column *Avail*).	Free up space first. Running out of space in the middle of the installation produces `No space left on device` / `Errno 28`.
Processor with AVX	Any normal PC or Mac from the last ten years has it. **Ubuntu:** `grep -o -m1 avx /proc/cpuinfo` prints `avx`.	Without AVX, TensorFlow stops with `Illegal instruction`. Use another computer, or a virtual machine with AVX enabled.
Internet	Open any web page.	Needed only during installation (TensorFlow alone is ≈600 MB). Behind a company proxy, ask your IT department.
Graphics card	Not needed.	SPADE runs entirely on the processor. Messages about CUDA/GPU can be ignored.

1.3 Terminal basics (Route B)

Route B is done by typing commands. If you have never used a terminal, read this page.

Opening the terminal

- **Windows:** Start menu → type *PowerShell* → click *Windows PowerShell*. To open it **as administrator**: right-click it → *Run as administrator*.
- **Ubuntu:** `Ctrl`+`Alt`+`T`, or search for *Terminal*.
- **Mac:** `⌘`+`Space` → type *Terminal* → Enter.

Reading the command blocks

```
python --version   # this is a comment
```

- Copy **only the left part**; everything after `#` is an explanation (pasting it does no harm).
- One line = one command. Paste a line, press `Enter`, wait until the terminal shows the prompt again, then do the next line.
- Paste: `Ctrl`+`V` (Windows), `Ctrl`+`Shift`+`V` (Ubuntu), `⌘`+`V` (Mac).

How do I know a command finished?

The terminal shows its prompt again, ready for input: `PS C:\dev>` on Windows, `user@pc:~$` on Ubuntu, `user@Mac ~ %` on Mac. Long commands (installing) print many lines; wait for the prompt.

To stop a running command (for example SPADE itself): `Ctrl`+`C`.

The current folder

Commands act on the folder the terminal is "in". `cd FOLDER` enters a folder. Many errors such as "*No such file or directory*" or "*can't open file*" mean you are in the wrong folder: run `cd` to the SPADE folder and try again.

What is a virtual environment (`.venv`)?

A folder inside SPADE with its own Python and its own libraries, isolated from the rest of the computer. SPADE cannot break other programs and vice versa. You must **activate** it in every new terminal before using SPADE; when active, the prompt starts with `(.venv)`.

Passwords (Linux and Mac)

Commands starting with `sudo` ask for **your user password**. While you type it **nothing appears** (not even asterisks). That is normal: type it and press Enter. Your user must be an administrator.

How each step is presented in this guide

- **What it does**: why the step exists.
- The **commands** to run.
- ✓ **You should see**: the expected result. If you see it, go to the next step.
- × **If it fails**: the error messages that can appear *at that step*, with their cause and fix. Section 8 lists them all again.

SECTION 2 · ROUTE A

Windows installer (.exe)

Windows 10 / 11, 64-bit. No Python, Git or administrator rights. The installer contains everything.

Prerequisites

Only the checks in section 1.2: 64-bit Windows 10 (21H2+) or 11, ≈2 GB free on `C:`, and internet for the ≈290 MB download. Nothing else needs to be installed: the installer brings its own Python, all libraries (TensorFlow, Streamlit, fmpy…) and the OpenModelica simulation models (FMUs) for Windows.

Step A1 · Download the installer

What it does: gets the single file that contains SPADE.

1. Open https://github.com/lucianoradrigan/SPADE/releases in your browser.
2. Find the newest release called **"SPADE for Windows app-v…"**, marked *Latest* (Figure 1). Its title says which commit of SPADE it contains.
3. Scroll down to **Assets** and click the file `SPADE-Setup-<version>-<commit>.exe` (Figure 2). **Do not** download "Source code (zip)": that is not an installer.

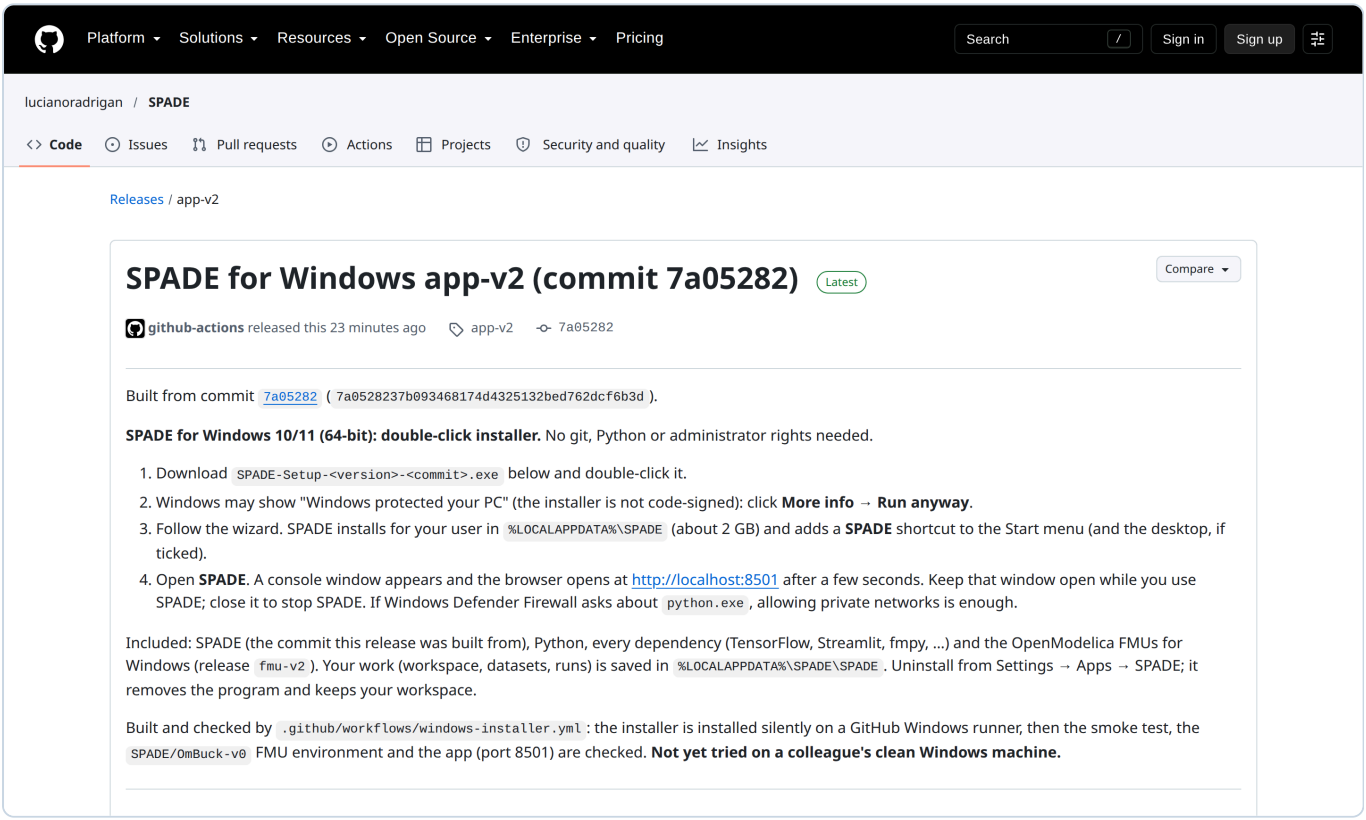


Figure 1. The release page on GitHub. The green *Latest* label marks the newest version.



Figure 2. The *Assets* list at the bottom of the release. Click the `.exe` (first line).

✓ **You should see:** the file `SPADE-Setup-…exe` (≈290 MB) in your *Downloads* folder.

> **× If it fails**
>
> - **The browser blocks the download** ("…isn't commonly downloaded", "…may be dangerous"). The file is not code-signed, so browsers may warn. In Microsoft Edge: open the downloads panel → **…** next to the file → **Keep** → **Show more** → **Keep anyway**. In Chrome: **Keep**.
> - **The download stops or the file is smaller than ≈290 MB.** The connection was interrupted: delete the file and download it again.

Step A2 · Run the installer

What it does: copies SPADE to your user folder and creates the shortcuts.

1. Double-click the downloaded `.exe`.
2. Windows may show a blue window **"Windows protected your PC"** (Microsoft Defender SmartScreen). This appears because the installer is not code-signed. Click **More info**, then **Run anyway**.
3. Follow the wizard. On the shortcuts page you can tick **Create a desktop shortcut**. Click **Install** and wait for the copy to finish (≈2 GB), then **Finish**.

✓ **You should see:** a **SPADE** entry in the Start menu (and on the desktop if you ticked it). SPADE is installed in `%LOCALAPPDATA%\SPADE`, only for your user, so no administrator password is requested.

> **× If it fails**
>
> - **There is no "Run anyway" button.** Click **More info** first: the button appears after it. If your organisation's policy hides it, the computer is managed and installing unsigned programs is blocked: ask your IT department, or use Route B.
> - **The wizard says the program needs a 64-bit Windows or a newer Windows version.** The installer accepts only 64-bit Windows 10 or later (section 1.2); use another computer, or Route B is not possible either.
> - **"There is not enough space on the disk".** Free ≈2 GB on `C:` and run the installer again.

Step A3 · Start SPADE

What it does: starts the SPADE application and its background worker (the process that runs long jobs), then opens SPADE in your browser.

1. Start menu → **SPADE** (or the desktop shortcut).
2. A black console window opens and prints:

```
SPADE 0.1.0 (commit 7a05282)
SPADE is starting. The browser opens at http://localhost:8501 in a few seconds.
Keep this window open while you use SPADE; close it to stop SPADE.
```

3. After a few seconds your browser opens SPADE (Figure 7, section 6).

✓ **You should see:** the *Workspace overview* page at `http://localhost:8501`.

> **Keep the black window open**
>
> It *is* SPADE. Closing it stops SPADE (and its worker). Starting the shortcut again while SPADE is running does not start a second copy: it prints "SPADE is already running" and just opens the browser.

× If it fails

- **Windows Defender Firewall asks about `python.exe`.** Allow access on **private networks**. That is enough: SPADE is only used from your own computer.
- **The browser does not open.** Open it yourself and go to `http://localhost:8501`. The first start can take longer while Windows scans the new files.
- **The browser shows a different program, or "This site can't be reached".** Another program is using port 8501, so SPADE thinks it is already running. Close the other program (or restart the computer) and start SPADE again.
- **The black window closes immediately.** Open PowerShell and run `& "$env:LOCALAPPDATA\SPADE\python\python.exe" "$env:LOCALAPPDATA\SPADE\SPADE\packaging\windows\launch.py"`: the window then stays open with the error message, which you can report.

Where things are, updating and uninstalling

Item	Details
Program	`%LOCALAPPDATA%\SPADE` (paste this into the File Explorer address bar to open it)
Your work	Workspace, datasets and runs are saved in `%LOCALAPPDATA%\SPADE\SPADE`
Update	Download and run the `.exe` of a newer release (step A1)
Uninstall	Settings → Apps → **SPADE** → Uninstall. The program is removed; your workspace is kept.

Verification status

The installer is built and tested automatically on a GitHub Windows machine (silent install, smoke test, the FMU environment and the app on port 8501). It has **not yet been tried on a user's clean Windows PC**, and the Windows dialogs above are described, not pictured (no Windows machine was available to capture them).

SECTION 3 · ROUTE B

Windows 10 / 11 from the repository

Everything is done in **PowerShell** (not WSL, not the old "Command Prompt").

Overview

Part 1 · Prerequisites: P1 winget · P2 Python 3.12 · P3 Git · P4 long paths (+ P5 Visual C++ only if needed).
Part 2 · Install SPADE: W1 download the code · W2 create the environment · W3 install SPADE · W4 smoke test · W5 open SPADE · W6 next time.

Part 1 · Prerequisites

P1 winget (Windows package manager)

What it is	A tool built into Windows that installs programs with one command. It is the easiest way to get Python and Git.
Do I have it?	`winget --version` → prints a version such as `v1.9.x`.
Install	It ships with Windows 11 and updated Windows 10. If missing: open the Microsoft Store, search **App Installer**, click *Get/Update*, then reopen PowerShell. **Without winget** you can still follow this guide using the "download from the website" option in P2 and P3.
If it fails	`winget : The term 'winget' is not recognized` → App Installer missing (see above), or use the website options. `No package found matching input criteria` → run `winget source update` and try again.

P2 Python 3.12, 64-bit

What it is	The programming language SPADE is written in.
Why this version	SPADE accepts **3.11 or 3.12** only. TensorFlow (the AI library) has no package for Python 3.13+ or for 32-bit Python, so those fail during installation.
Do I have it?	`py --list` → lists the installed Pythons; look for `3.12` (or `3.11`). "py is not recognized" means no Python from python.org/winget is installed.
Install	**Option 1 (winget):** `winget install --id Python.Python.3.12 -e` **Option 2 (website):** python.org → Downloads → Windows → choose a **Python 3.12** release → **Windows installer (64-bit)**. On the first screen tick **"Add python.exe to PATH"**, then *Install Now*.
Check	**Close PowerShell and open a new one** (it only finds new programs after reopening), then `py -3.12 --version` → `Python 3.12.x`.
If it fails	`py is not recognized` after reopening → reinstall with option 2 and make sure the "py launcher" option is ticked. `'python' is not recognized` or the **Microsoft Store opens** when typing `python` → that is a Windows shortcut, not Python. Always use `py -3.12` in this guide, or turn off Settings → Apps → Advanced app settings → *App execution aliases* → python. `No suitable Python runtime found` → 3.12 is not installed; repeat the install.

P3 Git

What it is	The program that downloads SPADE's code from GitHub and updates it later.
Do I have it?	`git --version` → `git version 2.x`.
Install	**Option 1:** `winget install --id Git.Git -e` **Option 2:** git-scm.com → *Download for Windows* → run it and accept every default option.
Check	Reopen PowerShell, then `git --version`.
If it fails	`git is not recognized` → PowerShell was not reopened after installing; reopen it. If it persists, restart the computer.

P4 Long paths enabled (once, as administrator)

What it is	A Windows setting. By default Windows refuses file paths longer than 260 characters.
Why	TensorFlow contains files with very long paths. Without this setting the installation in W3 fails with `WinError 206`.
Do I have it?	`(Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Control\FileSystem").LongPathsEnabled` → `1` means already enabled; skip P4.
Enable	Open PowerShell **as administrator** (right-click → *Run as administrator* → Yes), run: `New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\FileSystem" -Name "LongPathsEnabled" -Value 1 -PropertyType DWORD -Force` then **restart the computer**.
You should see	a few lines ending with `LongPathsEnabled : 1`.
If it fails	`Requested registry access is not allowed` → PowerShell is not running as administrator. Close it and open it with *Run as administrator*. If you have no administrator account, ask your IT department, or use Route A.

P5 Microsoft Visual C++ Redistributable (only if needed)

What it is	Standard Microsoft system libraries used by many programs, including TensorFlow. Most computers already have them.
When	Only if W4 or W5 shows `DLL load failed while importing _pywrap_tensorflow_internal`.
Do I have it?	Settings → Apps → search *Visual C++* → "Microsoft Visual C++ 2015-2022 Redistributable (x64)".
Install	`winget install Microsoft.VCRedist.2015+.x64`, then reopen PowerShell.

Part 2 · Install SPADE

Use a **normal** PowerShell (not administrator) from here on.

W1

Download the code into a short path

What it does: creates `C:\dev` and copies the SPADE repository into `C:\dev\SPADE`. A short path helps with Windows' path-length limit.

```
mkdir C:\dev -Force
cd C:\dev
git clone https://github.com/lucianoradrigan/SPADE.git
cd SPADE
```

✓ **You should see:** "Cloning into 'SPADE'...", progress lines, then the prompt `PS C:\dev\SPADE>`.

× If it fails

- `destination path 'SPADE' already exists` → you cloned it before. Run `cd SPADE` and `git pull` instead.
- `Could not resolve host: github.com` → no internet connection, or a proxy blocks it.
- `git is not recognized` → back to P3.

W2

Create and activate the environment

What it does: creates the isolated `.venv` folder with its own Python 3.12, then activates it in this window.

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
```

✓ **You should see:** no message after the first line; after the second, the prompt starts with `(.venv)`.

× If it fails

- `Activate.ps1 cannot be loaded because running scripts is disabled on this system` → PowerShell blocks scripts by default. Allow them for your user only, once: `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned` (answer **Y** if asked), then run the activate line again.
- `No suitable Python runtime found` → Python 3.12 is missing (P2).

W3

Install SPADE and its libraries

(several minutes)

What it does: updates pip (the Python package installer), then downloads and installs SPADE with every library it needs. `dev` adds the testing tools; `viz` adds the graphical interface (Streamlit, Plotly). Both are needed.

```
python -m pip install --upgrade pip
pip install -e ".[dev,viz]"
```

✓ **You should see:** many "Collecting..." and "Downloading..." lines (TensorFlow is the largest, ≈600 MB), ending with `Successfully installed …` and a long list of packages.

× If it fails

- `OSError: [Errno 2] No such file or directory: …tensorflow\include\…` or `WinError 206` → long paths are not enabled (P4), or the computer was not restarted after P4. Fix, restart, then run W2 activate + W3 again.
- `Could not find a version that satisfies the requirement tensorflow` → the environment was made with 32-bit Python or Python 3.13+. Delete the `.venv` folder and redo W2 with `py -3.12` (64-bit).
- `requires a different Python: 3.13.x not in '<3.13,>=3.11'` → same cause: recreate `.venv` with `py -3.12`.
- `ReadTimeoutError` / `Connection reset` → slow or interrupted connection. Run the same command again (already-downloaded packages are reused), or add `--default-timeout=100`.
- `No space left on device` / `Errno 28` → free ≈3 GB and run it again.
- Copy the command exactly, including the quotes around `".[dev,viz]"`.

W4

Check the installation (smoke test)

What it does: a ≈1-minute automatic check of the whole installation. Section 6 explains each of its checks.

```
python scripts\smoke_test.py
```

✓ **You should see:** five lines starting with `[PASS]` and the last line `6. no exceptions: PASS -- 5/5 checks passed` (Figure 6, section 6).

× If it fails

- `DLL load failed while importing _pywrap_tensorflow_internal` → install P5, reopen PowerShell, `cd C:\dev\SPADE`, activate (W2 second line), run W4 again.
- `can't open file … smoke_test.py` → you are not in the SPADE folder: `cd C:\dev\SPADE`.
- A line with `[FAIL]` → see "When a check fails" in section 6.

W5

Open SPADE

What it does: starts the SPADE web application on your computer and opens it in the browser.

```
streamlit run src\driveflow\viz\app.py
```

✓ **You should see:** "You can now view your Streamlit app in your browser. Local URL: http://localhost:8501", and the browser opens the Overview page (Figure 7). Leave PowerShell open; `Ctrl`+`C` in it stops SPADE.

× **If it fails**

- **Windows Defender Firewall asks about** `python.exe` → allowing **private networks** is enough.
- `streamlit is not recognized` → the environment is not active (no `(.venv)`): run `.\.venv\Scripts\Activate.ps1`, or use `python -m streamlit run src\driveflow\viz\app.py`.
- `Port 8501 is already in use` → SPADE (or another Streamlit app) is already running. Close it, or add `--server.port 8502` and open `http://localhost:8502`.
- Messages about **oneDNN**, **CUDA** or **GPU** → informational; ignore them.

W6

Every next time

Open PowerShell and run:

```
cd C:\dev\SPADE
.\.venv\Scripts\Activate.ps1
streamlit run src\driveflow\viz\app.py
```

Alternative: uv instead of pip

Advanced users can use `uv`: `powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"`, then `uv venv --python 3.12 .venv`, activate, and `uv pip install -e ".[dev,viz]"`.

Verification status

These Windows steps were checked by dependency resolution and the CI smoke test on a GitHub Windows machine; they have **not yet been run by hand on a real Windows PC**.

SECTION 4 · ROUTE B

Linux (Ubuntu 22.04 · 24.04 · 26.04)

Open the Terminal with `Ctrl`+`Alt`+`T`. Linux is the system SPADE is developed and tested on.

Overview

Part 1 · Prerequisites: P1 system tools (git, curl, build-essential) · P2 uv · P3 Python 3.11/3.12 (provided by uv).
Part 2 · Install SPADE: L1 download the code · L2 create the environment · L3 install SPADE · L4 smoke test · L5 open SPADE · L6 next time.

Part 1 · Prerequisites

P1 System tools: git, curl, build-essential

What they are	**git** downloads the code; **curl** downloads files from the terminal (used to install uv); **build-essential** is the C compiler set, needed by a few Python packages.
Do I have them?	`git --version`, `curl --version`, `gcc --version` → each prints a version.

Install	`sudo apt update`
	`sudo apt install -y git curl build-essential`
	(asks for your password; nothing is shown while typing).

If it fails	`Could not get lock /var/lib/dpkg/lock-frontend` → Ubuntu is installing updates in the background. Wait a few minutes and run it again. `user is not in the sudoers file` → your account is not an administrator; ask the computer's administrator. `Temporary failure resolving 'archive.ubuntu.com'` → no internet connection.

P2 uv (recommended)

What it is	A fast Python package installer that can also download the right Python version by itself.
Why	Each Ubuntu ships a different Python: **22.04 → 3.10 (too old)**, **24.04 → 3.12 (works)**, **26.04 → 3.14 (too new)**. With uv the same commands work on all three, without touching the system Python (which Ubuntu itself needs — never replace it).
Do I have it?	`uv --version` → e.g. `uv 0.9.x`.
Install	`curl -LsSf https://astral.sh/uv/install.sh | sh` `source "$HOME/.local/bin/env"`
Check	`uv --version`
If it fails	`uv: command not found` right after installing → the second line was skipped. Run `source "$HOME/.local/bin/env"`, or open a new terminal. `curl: command not found` → do P1 first.

P3 Python 3.11 or 3.12

With uv	Nothing to do: step L2 downloads Python 3.11 automatically.

Without uv	**Ubuntu 24.04:** the system Python 3.12 works; install its venv module: `sudo apt install -y python3.12-venv`.
	Ubuntu 22.04: add Python 3.11 next to the system one (it does not replace it):
	`sudo apt install -y software-properties-common`
	`sudo add-apt-repository -y ppa:deadsnakes/ppa`
	`sudo apt update`
	`sudo apt install -y python3.11 python3.11-venv`
	Ubuntu 26.04: use uv.

Check	`python3.12 --version` (or `python3.11 --version`).

Part 2 · Install SPADE

L1 Download the code

What it does: copies the SPADE repository into a `SPADE` folder inside your current folder (your home folder if you just opened the terminal).

```
git clone https://github.com/lucianoradrigan/SPADE.git
cd SPADE
```

✓ **You should see:** "Cloning into 'SPADE'...", then the prompt ends with `~/SPADE$`.

× If it fails

- `destination path 'SPADE' already exists` → already cloned: `cd SPADE` and `git pull`.
- `Could not resolve host: github.com` → no internet connection.

L2 Create and activate the environment

What it does: uv downloads Python 3.11 (if needed) and creates the isolated `.venv` folder; `source` activates it.

```
uv venv --python 3.11 .venv
source .venv/bin/activate
```

✓ **You should see:** "Using CPython 3.11.x ... Creating virtual environment at: .venv", then the prompt starts with `(.venv)`.

Without uv: `python3.12 -m venv .venv` (24.04) or `python3.11 -m venv .venv` (22.04 + deadsnakes), then `source .venv/bin/activate`.

× If it fails

- `The virtual environment was not created successfully because ensurepip is not available` → the venv module is missing: `sudo apt install python3.12-venv` (or `python3.11-venv`).
- `uv: command not found` → `source "$HOME/.local/bin/env"` (P2).

L3

Install SPADE and its libraries

(several minutes)

What it does: installs SPADE in editable mode with its testing tools (`dev`) and graphical interface (`viz`). Both extras are needed.

```
uv pip install -e ".[dev,viz]"
```

Without uv: `python -m pip install --upgrade pip`, then `pip install -e ".[dev,viz]"`.

✓ **You should see:** a list of resolved and installed packages ending without errors (with pip: `Successfully installed …`).

Exactly the verified versions (optional)

To install exactly the package set the tests were verified with (Python 3.11.15, TensorFlow 2.21.0, Streamlit 1.62.0), use instead: `uv pip install -r requirements-lock.txt` followed by `uv pip install -e . --no-deps`.

× If it fails

- `Package 'driveflow' requires a different Python: 3.10.12 not in '<3.13,>=3.11'` → the environment uses Ubuntu 22.04's system Python. Delete `.venv` (`rm -rf .venv`) and redo L2 with uv.
- Same message with `3.13` / `3.14` → newer Ubuntu's system Python: `rm -rf .venv`, then `uv venv --python 3.12 .venv`.
- Network timeouts → run the same command again.
- `No space left on device` → free ≈3 GB.

L4

Check the installation (smoke test)

```
python scripts/smoke_test.py
```

✓ **You should see:** five `[PASS]` lines and `6. no exceptions: PASS -- 5/5 checks passed` (Figure 6). `echo $?` right after prints `0`.

× If it fails

- `ModuleNotFoundError: No module named 'streamlit'` → installed without the `viz` extra: redo L3 with exactly `".[dev,viz]"`.
- `Illegal instruction (core dumped)` → the processor has no AVX (section 1.2).
- Messages `Could not find cuda drivers on your machine, GPU will not be used` / `Unable to register cuDNN factory` → informational; ignore them.
- A line with `[FAIL]` → section 6, "When a check fails".

L5

Open SPADE

```
streamlit run src/driveflow/viz/app.py
```

✓ **You should see:** "Local URL: http://localhost:8501" and the browser opens the Overview page (Figure 7). `Ctrl`+`C` in the terminal stops SPADE.

× If it fails

- `streamlit: command not found` → environment not active: `source .venv/bin/activate`, or use `python -m streamlit run src/driveflow/viz/app.py`.
- `Port 8501 is already in use` → close the other Streamlit, or add `--server.port 8502`.
- **The browser does not open** → open `http://localhost:8501` yourself.
- **Remote server without a screen** → add `--server.headless true --server.address 0.0.0.0`, open port 8501 in the firewall and browse to `http://SERVER-IP:8501`.
- MPC pages very slow → `export OMP_NUM_THREADS=1` before starting.

L6

Every next time

```
cd SPADE
source .venv/bin/activate
streamlit run src/driveflow/viz/app.py
```

Verification status

Executed on **Ubuntu 26.04** with Python 3.11 and 3.12 installed by uv. Ubuntu 22.04 and 24.04 were checked by dependency resolution and the CI job (Ubuntu 24.04), not by hand.

SECTION 5 · ROUTE B

macOS (Apple Silicon & Intel)

Open the Terminal: ⌘+Space → "Terminal".

Overview

Step 0: find out which Mac you have. **Part 1 · Prerequisites:** P1 Command Line Tools · P2 Homebrew · P3 Python 3.12 and Git.
Part 2 · Install SPADE: M1 download the code · M2 create the environment · M3 install SPADE · M4 smoke test · M5 open SPADE · M6 next time.

Step 0 · Which Mac do I have? (important)

```
uname -m      # arm64 = Apple Silicon (M1–M4) · x86_64 = Intel
```

Or: Apple menu → *About This Mac*: "Chip Apple M…" = Apple Silicon; "Processor … Intel" = Intel. If your Mac is Apple Silicon but `uname -m` prints `x86_64`, the Terminal is running under **Rosetta** (Intel emulation): in Finder → Applications → Utilities → Terminal → *Get Info*, untick **Open using Rosetta**, and reopen it.

	Apple Silicon (M1–M4)	Intel
Python	3.11 or 3.12	**3.12 only** (with 3.11 an old, incompatible TensorFlow is chosen)
TensorFlow	2.21 (current)	2.16.2 at most: Google publishes no newer build for Intel Macs
Extra step	None	Install with `-c constraints/macos-intel.txt` (M3), which fixes compatible versions of NumPy, pyarrow and Streamlit

Part 1 · Prerequisites

P1 Xcode Command Line Tools

What it is	Apple's developer tools: they include **Git** and the compilers some Python packages need. (Not the full Xcode app.)
Do I have it?	`xcode-select -p` → prints a path such as `/Library/Developer/CommandLineTools`.
Install	`xcode-select --install` → a window opens: click **Install**, accept the licence, wait for the download.
If it fails	`command line tools are already installed` → nothing to do. `xcrun: error: invalid active developer path` (later, when using git) → the tools are missing or broken: run `xcode-select --install`.

P2 Homebrew

What it is	The usual package manager for macOS; it installs Python and Git with one command.
Do I have it?	`brew --version` → `Homebrew 4.x`. If so, skip to P3.

Install	`/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
	It asks for your password and to press Enter. At the end it prints **"Next steps"** with two commands that add `brew` to your PATH. **Copy and run them.** On Apple Silicon they are:
	`echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile`
	`eval "$(/opt/homebrew/bin/brew shellenv)"`

If it fails	`zsh: command not found: brew` after installing → the "Next steps" commands were not run; run them (above), or open a new Terminal.

P3 Python 3.12 and Git

Why	SPADE needs Python 3.11/3.12 (3.12 on Intel). The `python3` that macOS or Homebrew provides by default may be another version, so this guide always calls `python3.12` explicitly.
Install	`brew install python@3.12 git`
Check	`python3.12 --version` → `Python 3.12.x` · `git --version`
If it fails	`command not found: python3.12` → Homebrew is not on the PATH (P2 "Next steps"), or call it by its full path: `/opt/homebrew/bin/python3.12` (Apple Silicon) or `/usr/local/bin/python3.12` (Intel).

Part 2 · Install SPADE

M1 Download the code

```
git clone https://github.com/lucianoradrigan/SPADE.git
cd SPADE
```

✓ **You should see:** "Cloning into 'SPADE'...", then the prompt shows `SPADE`.

× If it fails

- `destination path 'SPADE' already exists` → `cd SPADE` and `git pull`.
- `xcrun: error: invalid active developer path` → P1.

M2 Create and activate the environment

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
```

✓ **You should see:** the prompt starts with `(.venv)`; pip reports "Successfully installed pip-...".

M3

Install SPADE and its libraries

— choose the line for your Mac (Step 0)

Apple Silicon:

```
pip install -e ".[dev,viz]"
```

Intel (the `-c` part is mandatory):

```
pip install -e ".[dev,viz]" -c constraints/macos-intel.txt
```

✓ **You should see:** `Successfully installed …` with a long list of packages.

× If it fails

- `No matching distribution found for tensorflow==2.21…` → you are on an Intel Mac (or a Rosetta Terminal) without the constraints. Use the Intel line; on Apple Silicon, fix Rosetta (Step 0) and recreate `.venv`.
- `requires a different Python: 3.13.x` → the environment was created with another Python: `rm -rf .venv` and redo M2 with `python3.12`.
- `zsh: no matches found: .[dev,viz]` → the quotes were left out; macOS's shell (zsh) needs them. Copy the line exactly: `".[dev,viz]"`.

M4

Check the installation (smoke test)

```
python scripts/smoke_test.py
```

✓ **You should see:** five `[PASS]` lines and `6. no exceptions: PASS -- 5/5 checks passed` (Figure 6). The very first run can be slow: macOS (Gatekeeper) scans the newly installed files once.

× If it fails

- `ImportError: pyarrow requires NumPy 2.0 or newer, found 1.26.4` → Intel Mac installed without constraints: run the Intel line of M3.
- Errors mentioning `keras` 2.x → Intel Mac with Python 3.11: `rm -rf .venv`, redo M2 with `python3.12`, then the Intel line of M3.
- `zsh: illegal hardware instruction python` → the environment was created under Rosetta: `rm -rf .venv` and redo M2 with `/opt/homebrew/bin/python3.12` in a native Terminal.

M5

Open SPADE

```
streamlit run src/driveflow/viz/app.py
```

✓ **You should see:** the browser opens at `http://localhost:8501` (Figure 7). If macOS asks whether "Python" may accept incoming network connections, you can click **Deny** for local use. `Ctrl`+`C` stops SPADE.

× If it fails

- `streamlit: command not found` → `source .venv/bin/activate`, or `python -m streamlit run src/driveflow/viz/app.py`.
- `Port 8501 is already in use` → add `--server.port 8502`.

M6

Every next time:

`cd SPADE` → `source .venv/bin/activate` → `streamlit run src/driveflow/viz/app.py`

Do not install `tensorflow-metal`

SPADE does not need it (everything runs on the CPU) and it was not tested with it.

Verification status

Checked by dependency resolution for both architectures, by the CI smoke test on an Apple Silicon GitHub machine, and by emulating the Intel package set on Linux; **not yet run by hand on a real Mac**.

SECTION 6

Verifying the installation

What the smoke test checks, what a correct result looks like, and a first simulation by hand.

6.1 The smoke test, check by check

#	Check	What it verifies	If this line says [FAIL]
1	Dependencies import	Every library loads; their versions are printed.	A library is missing or incompatible: repeat the "install SPADE" step of your system, exactly as written (with `".[dev,viz]"`; on Intel Mac with the constraints).
2	Application starts	The Overview page renders with no error.	Usually the `viz` extra is missing: same fix as check 1.
3	Modules load	Simulation, control, data generation, AI registry, monitoring and rule files load.	Files of the repository are missing: you may be in the wrong folder, or the clone was interrupted. Run `git status` in the SPADE folder; if in doubt, clone again.
4	Short simulations	DC motor (PI and MPC), VSC with DPC and PMSM FOC run briefly; outputs must be finite.	Report it (see "Reporting a problem", section 8).
5	Inference	The registered classifier gives a prediction on a freshly simulated window.	Report it.
6	No exceptions	Summary of checks 1–5.	Look at which earlier line failed.

Optional: `python scripts/smoke_test.py --launch` additionally starts the real SPADE server and checks it answers.

```
(.venv) $ python scripts/smoke_test.py
SPADE smoke test -- Python 3.11.15 on Linux 7.0.0-38-generic (x86_64)
[PASS] 1. dependencies import (2.3 s) numpy 2.4.6; scipy 1.17.1; pandas 3.0.5; pyarrow 25.0.1; sklearn 1.9.
0; yaml 6.0.3; gymnasium 1.3.0; threadpoolctl 3.6.0; tensorflow 2.21.0; keras 3.15.1; streamlit 1.62.0; plot
ly 7.0.0
[PASS] 2. application starts (Overview page renders) (0.8 s) AppTest: no exceptions
[PASS] 3. platform modules load (0.0 s) rule sets: ['vsc_dpc']
[PASS] 4. short simulations (DC+PI, DC+MPC, VSC+DPC, PMSM FOC) (0.5 s) final speed 470.3 rpm, VSC |vc| 50.2
V
[PASS] 5. inference with a registered model (0.7 s) pc_server_5class_w2048/2026-10-20_run01 -> uncertain (0.
28, calibrated)
6. no exceptions: PASS -- 5/5 checks passed
```

Figure 6. Real smoke-test output on Linux (TensorFlow's informational log lines removed). Versions, paths and timings on your computer will differ; what matters is five `[PASS]` lines and `5/5 checks passed`.

6.2 SPADE in the browser

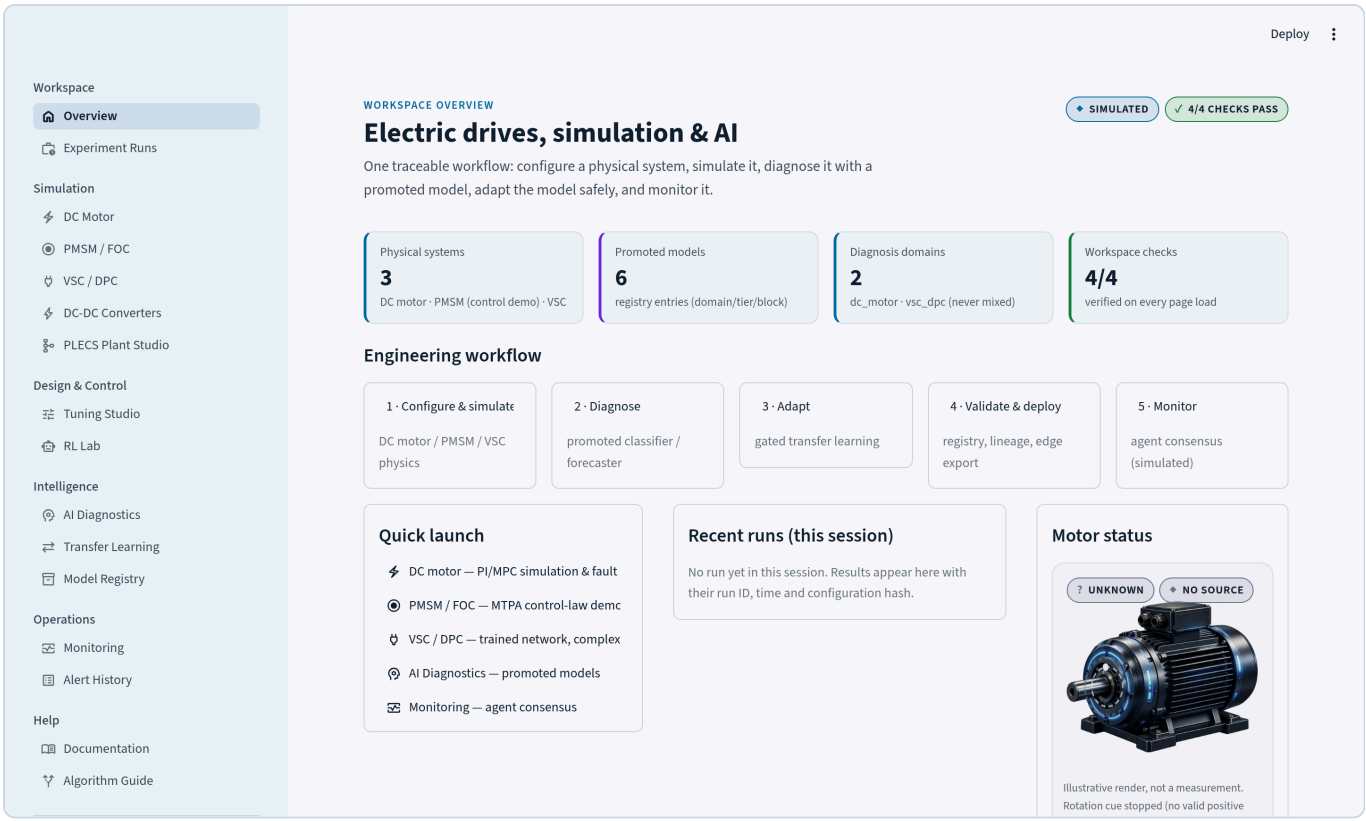


Figure 7. The Overview page at `http://localhost:8501`. Left: navigation by group. Top right: *4/4 CHECKS PASS* confirms the workspace is complete. "No run yet" and motor status "UNKNOWN" are normal on a fresh start.

6.3 A first simulation by hand

1. In the left menu, under **Simulation**, click **DC Motor**.
2. Leave the default parameters and click **Run simulation** (below the parameters on the left).
3. After a few seconds, plots of torque, armature voltage and vibration appear, followed by a **Run metadata** table with a Run ID such as `dc-001` (Figure 8). The button turns grey: the shown result already belongs to this exact configuration.

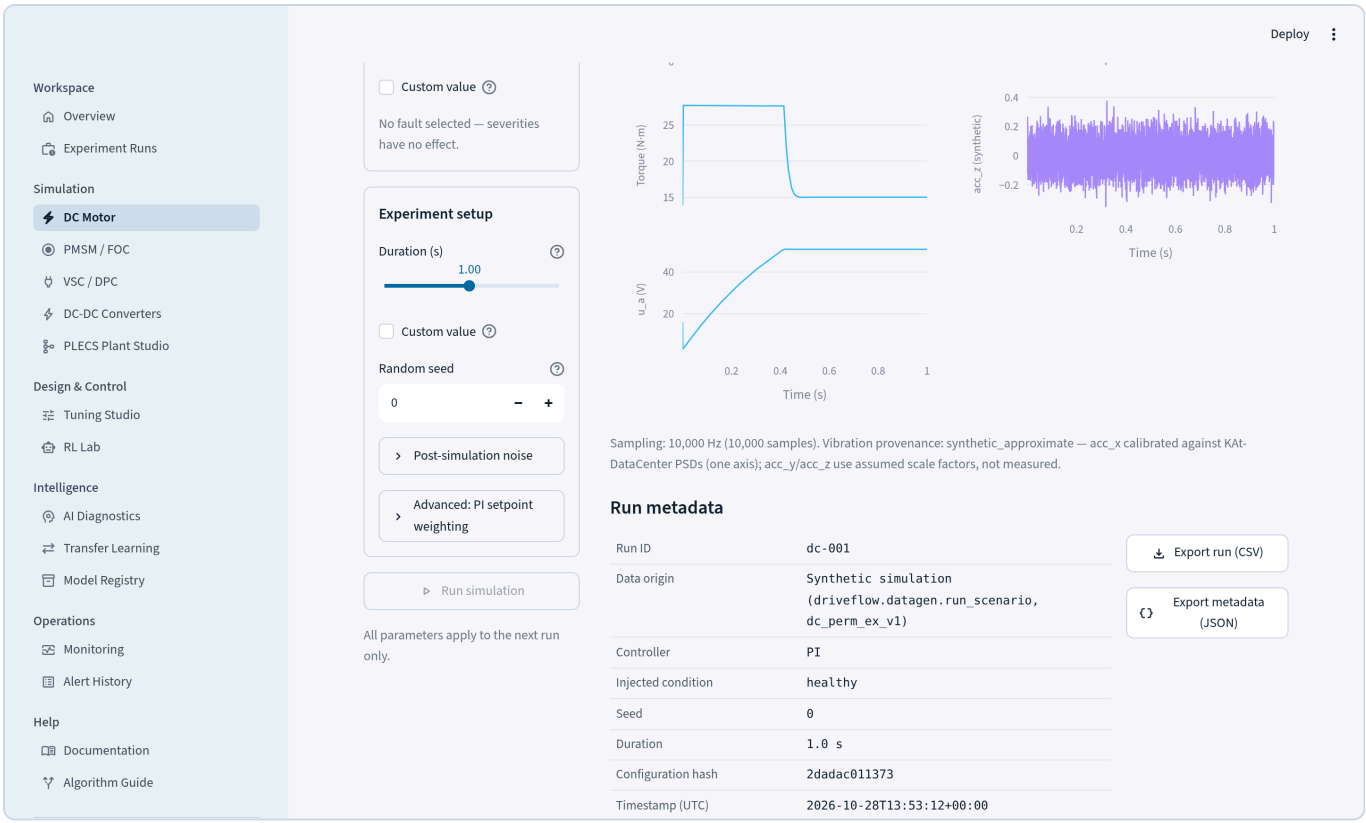


Figure 8. DC Motor after *Run simulation*: signal plots and the run metadata (Run ID, data origin "Synthetic simulation", controller, injected condition, configuration hash). If you see this, SPADE is installed and working.

× If it fails

- **A red error box appears** instead of plots → copy its text and report it (section 8).
- **Nothing happens for a long time** → look at the terminal (or the black window on Route A) for an error message.

SECTION 7

After installing

Daily use, the background worker, updating and uninstalling.

7.1 Opening SPADE every day

Route / system	What to do
Route A	Start menu → **SPADE**. Close the black window to stop it.
Windows (B)	`cd C:\dev\SPADE` → `.\.venv\Scripts\Activate.ps1` → `streamlit run src\driveflow\viz\app.py`
Linux / Mac (B)	`cd SPADE` → `source .venv/bin/activate` → `streamlit run src/driveflow/viz/app.py`

The legacy command `streamlit run src/driveflow/viz/dashboard.py` opens the same app and still works.

7.2 Long jobs and the worker

Sweeps, tuning, reinforcement-learning training and classifier training run as **jobs** in a separate background process, the **worker**, so they keep running while you move between pages.

- **Route A:** the SPADE shortcut starts the worker automatically.
- **Route B:** start it from **Experiment Runs → Start local worker**, or in a second terminal with the environment active: `python -m driveflow.jobs.worker serve`.

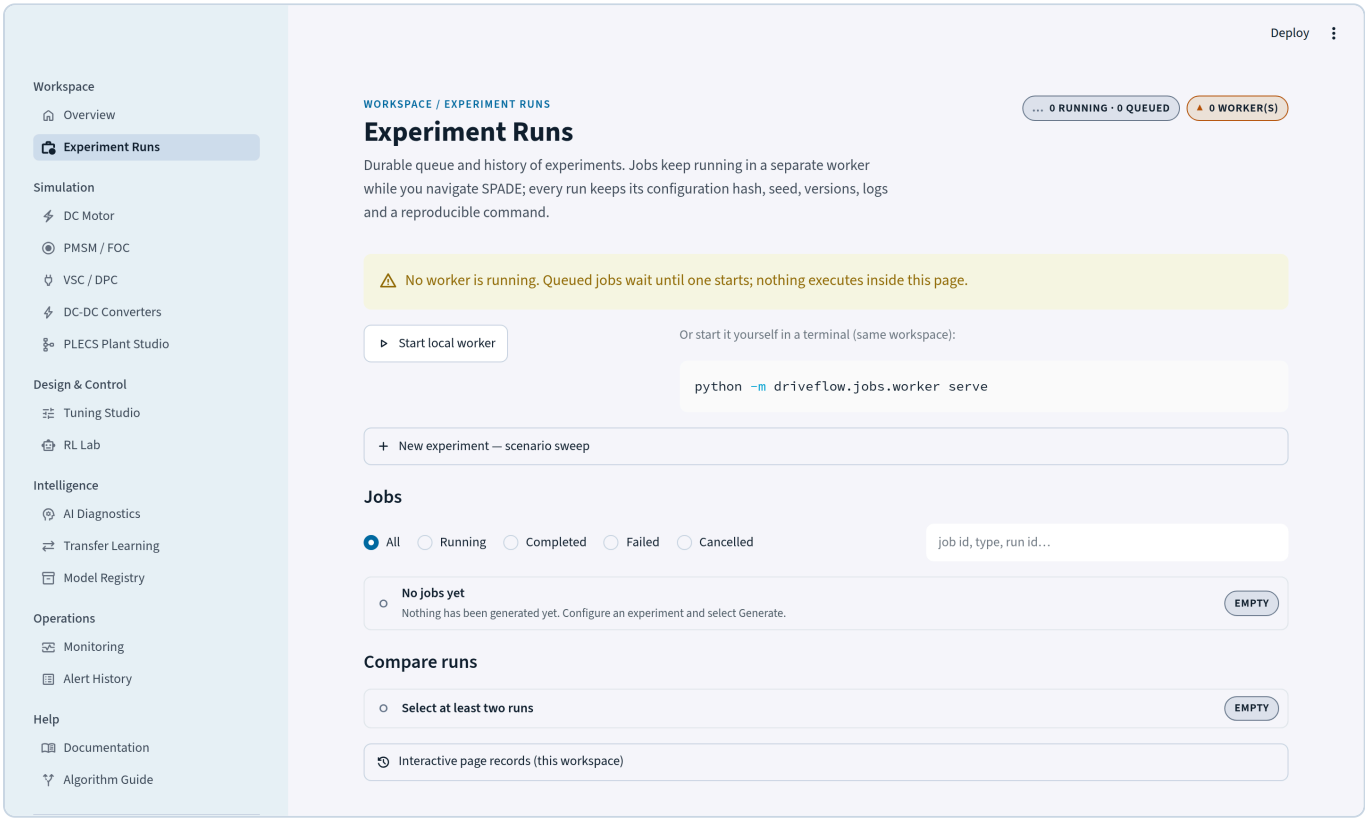


Figure 9. *Workspace → Experiment Runs* with no worker: the yellow notice and the *0 WORKER(S)* badge. Click **Start local worker** or run the command shown. The sidebar also shows "Jobs: N running · N queued · N worker(s)" on every page.

× If it fails

- **A job stays "queued" forever** → no worker is running: start it as above.

7.3 Updating (Route B)

Inside the SPADE folder, with the environment active:

```
git pull
pip install -e ".[dev,viz]"          # Linux with uv: uv pip install -e ".[dev,viz]"
# Intel Mac: add  -c constraints/macos-intel.txt
python scripts/smoke_test.py
```

× If it fails

- `Your local changes … would be overwritten by merge` → you edited files of the repository. Save your changes elsewhere, then `git stash` and `git pull` again (or ask a developer).

7.4 Uninstalling

- **Route A:** Settings → Apps → SPADE → Uninstall (your workspace is kept).
- **Route B:** delete the `SPADE` folder (it contains `.venv` and your workspace). Python, Git, uv and Homebrew can stay, or be removed like any other program.

7.5 Useful settings

- **Light or dark theme:** ⋮ menu (top right) → Settings → Theme. By default it follows your system.
- **Hide TensorFlow's informational messages:** Linux/Mac `export TF_CPP_MIN_LOG_LEVEL=2` · Windows `$env:TF_CPP_MIN_LOG_LEVEL = "2"` (before starting SPADE).

About the data

Everything SPADE shows is **simulated**: there is no connection to real hardware. Pages say so with the *SIMULATED* label.

SECTION 8

Error index

Every error in this guide, in one place. Search for a few words of your message. "Step" tells you where it is explained.

Message (or part of it)	Sys.	Cause	Fix · step
`running scripts is disabled on this system`	Win	PowerShell execution policy	`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` · W2
`WinError 206` / `No such file or directory …tensorflow\include…`	Win	260-character path limit	Enable long paths, restart, short path `C:\dev` · P4, W3
`Requested registry access is not allowed`	Win	Not running as administrator	Run PowerShell as administrator · P4
`Could not find a version that satisfies the requirement tensorflow`	Win	32-bit Python or 3.13+	Recreate `.venv` with 64-bit `py -3.12` · P2, W3
`DLL load failed while importing _pywrap_tensorflow_internal`	Win	Missing Visual C++ Redistributable	`winget install Microsoft.VCRedist.2015+.x64` · P5
`'python' is not recognized` / Microsoft Store opens	Win	Windows alias, not Python	Use `py -3.12`; disable app-execution aliases · P2
`winget … is not recognized`	Win	App Installer missing	Microsoft Store → App Installer, or website downloads · P1
"Windows protected your PC"	Win	Installer not code-signed	More info → Run anyway · A2
"…isn't commonly downloaded" (browser)	Win	Unsigned download	Keep → Keep anyway · A1
`requires a different Python: 3.10 / 3.13 / 3.14 …`	All	Unsupported Python in `.venv`	Delete `.venv`, recreate with 3.11/3.12 · W3, L3, M3
`ensurepip is not available`	Linux	venv module missing	`sudo apt install python3.12-venv` · L2
`Could not get lock /var/lib/dpkg/lock-frontend`	Linux	Updates running in the background	Wait, retry · P1
`is not in the sudoers file`	Linux	Not an administrator	Ask the administrator · P1
`uv: command not found`	Linux	PATH not refreshed	`source "$HOME/.local/bin/env"` · P2
`Illegal instruction (core dumped)`	All	Processor without AVX	Another computer / VM with AVX · 1.2
`command not found: brew`	Mac	Homebrew "Next steps" not run	Run the `brew shellenv` lines · P2
`command not found: python3.12`	Mac	Homebrew not on PATH	P2 "Next steps", or full path · P3
`xcrun: error: invalid active developer path`	Mac	Command Line Tools missing	`xcode-select --install` · P1
`No matching distribution found for tensorflow==2.21`	Mac	Intel Mac / Rosetta without constraints	Intel line of M3; fix Rosetta · Step 0, M3
`zsh: no matches found: .[dev,viz]`	Mac	Quotes left out	Use `".[dev,viz]"` with quotes · M3
`pyarrow requires NumPy 2.0 or newer`	Mac	Intel Mac without constraints	`-c constraints/macos-intel.txt` · M3
errors mentioning `keras` 2.x	Mac	Python 3.11 on Intel Mac	Recreate with 3.12 + constraints · M4

`illegal hardware instruction`	Mac	Environment created under Rosetta	Recreate with `/opt/homebrew/bin/python3.12` · M4
`destination path 'SPADE' already exists`	All	Already cloned	`cd SPADE`, `git pull` · W1, L1, M1
`Could not resolve host: github.com`	All	No internet / proxy	Check connection · W1, L1
`ReadTimeoutError` / connection reset	All	Slow network	Rerun; `--default-timeout=100` · W3
`No space left on device` / `Errno 28`	All	Disk full	Free ≈3 GB · 1.2
`No module named 'streamlit'`	All	Installed without `viz`	`pip install -e ".[dev,viz]"` · L4
`streamlit: command not found` / not recognized	All	Environment not active	Activate `.venv` or `python -m streamlit …` · W5, L5, M5
`can't open file … smoke_test.py`	All	Wrong folder	`cd` into SPADE · W4
`Port 8501 is already in use`	All	SPADE already open	Close it or `--server.port 8502` · W5
CUDA / cuDNN / GPU / oneDNN messages	All	Informational TensorFlow log	Ignore · L4

Reporting a problem

If nothing matches, collect: (1) your operating system and version, (2) the route and the step where it failed, (3) the full error message (the last ≈20 lines of the terminal), and (4) the output of `python scripts/smoke_test.py` if you got that far. Send them to the SPADE maintainers.

APPENDIX

Glossary and quick summary

The technical words in this guide, one line each.

Term	Meaning
Repository (repo)	The project with all its source code, hosted on GitHub.
Git / clone / pull	Git downloads and updates code. "Clone" = download a copy; "pull" = update it.
Python	The programming language SPADE is written in.
pip / uv	Programs that install Python libraries. `uv` is faster and can also install Python itself.
Virtual environment (.venv)	An isolated folder with SPADE's Python and libraries; "activate" it before using SPADE.
Extras `[dev,viz]`	Optional parts of SPADE: `dev` = testing tools, `viz` = graphical interface.
winget / Homebrew / apt	Program installers for Windows, macOS and Ubuntu.
PATH	The list of folders where the system looks for the programs you type.
Administrator / sudo	An account allowed to change system settings; `sudo` runs one command as administrator.
Streamlit	The library that shows SPADE as a web page in your browser.
localhost:8501	Your own computer, port 8501. Nothing goes out to the internet.
TensorFlow	The AI library behind SPADE's diagnosis models; the heaviest part of the install.
AVX	A processor feature TensorFlow requires.
Rosetta	macOS emulation of Intel programs on Apple Silicon; must be off for SPADE.
Smoke test	The quick automatic check `scripts/smoke_test.py` (section 6).
Worker / job	The background process that runs long tasks ("jobs").
FMU	A packaged simulation model (FMI standard), here exported from OpenModelica.

Summary

	Windows · .exe	Windows · repo	Linux · repo	macOS · repo
Prerequisites	None	winget, Python 3.12, Git, long paths	git, curl, build-essential, uv	Command Line Tools, Homebrew, Python 3.12, Git
Install	Double-click the .exe	`pip install -e ".[dev,viz]"`	`uv pip install -e ".[dev,viz]"`	`pip install -e ".[dev,viz]"` (+ `-c constraints/macos-intel.txt` on Intel)
Check	Browser opens SPADE	`python scripts/smoke_test.py` → 5/5 checks passed		

Open	Start menu → SPADE	`streamlit run src/driveflow/viz/app.py` → `http://localhost:8501`		
Run by hand on a real machine	GitHub runner only	No (CI only)	Yes (Ubuntu 26.04)	No (CI + emulation)